| | Result | Size | Time | Cycles | GPU | SM Frequency | Process | Attributes |
|---|---|---|---|---|---|---|---|---|
| Current | 599 - dir24_8_loo | (782, 1, 1)x(256, 1, 1) | 13,02 us | 19 281 | 0 - NVIDIA GeForce RTX 3050 Ti Laptop GPU | 1,48 Ghz | [22996] gpu_dir24_8_lpm_REAL_TRAFFIC_V1.exe | |

Summary | Details | Source | Context | Comments | Raw | Session

Compare | Tools | View | Export

## Command line profiler metrics

| | | | |
|---|---|---|---|
| dram__throughput.avg.pct_of_peak_sustained_elapsed [%] | 64,24 | gpu__time_duration.sum [us] | 13,02 |
| lts__t_sector_hit_rate.pct [%] | 52,26 | sm__throughput.avg.pct_of_peak_sustained_elapsed [%] | 19,97 |
| smsp__warp_issue_stalled_long_scoreboard_per_warp_active.pct [%] | 100,64 | - | - |

## GPU Speed Of Light Throughput

High-level overview of the throughput for compute and memory resources of the GPU. For each unit, the throughput reports the achieved percentage of utilization with respect to the theoretical maximum. Breakdowns show the throughput for each individual sub-metric of Compute and Memory to clearly identify the highest contributor.

| | | | |
|---|---|---|---|
| Compute (SM) Throughput [%] | 19,97 | Duration [us] | 13,02 |
| Memory Throughput [%] | 64,24 | Elapsed Cycles [cycle] | 19 281 |
| L1/TEX Cache Throughput [%] | 22,03 | SM Active Cycles [cycle] | 17 028,40 |
| L2 Cache Throughput [%] | 19,40 | SM Frequency [Ghz] | 1,48 |
| DRAM Throughput [%] | 64,24 | DRAM Frequency [Ghz] | 6,00 |

**High Memory Throughput** Memory is more heavily utilized than Compute: Look at the Memory Workload Analysis section to identify the DRAM bottleneck. Check memory replay (coalescing) metrics to make sure you're efficiently utilizing the bytes transferred. Also consider whether it is possible to do more work per memory access (kernel fusion) or whether there are values you can (re)compute.

Key Performance Indicators

## Launch Statistics

Summary of the configuration used to launch the kernel. The launch configuration defines the size of the kernel grid, the division of the grid into blocks, and the GPU resources needed to execute the kernel. Choosing an efficient launch configuration maximizes device utilization.

| | | | |
|---|---|---|---|
| Grid Size | 782 | Function Cache Configuration | CachePreferNone |
| Registers Per Thread [register/thread] | 16 | Static Shared Memory Per Block [byte/block] | 0 |
| Block Size | 256 | Dynamic Shared Memory Per Block [byte/block] | 0 |
| Threads [thread] | 200 192 | Driver Shared Memory Per Block [Kbyte/block] | 1,02 |
| Waves Per SM | 6,52 | Shared Memory Configuration Size [Kbyte] | 8,19 |
| Uses Green Context | 0 | Stack Size | 1 024 |
| # SMs [SM] | 20 | # TPCs | 10 |
| Enabled TPC IDs | all | - | - |

## Occupancy

Occupancy is the ratio of the number of active warps per multiprocessor to the maximum number of possible active warps. Another way to view occupancy is the percentage of the hardware's ability to process warps that is actively in use. Higher occupancy does not always result in higher performance, however, low occupancy always reduces the ability to hide latencies, resulting in overall performance degradation. Large discrepancies between the theoretical and the achieved occupancy during execution typically indicates highly imbalanced workloads.

| | | | |
|---|---|---|---|
| Theoretical Occupancy [%] | 100 | Block Limit Registers [block] | 16 |
| Theoretical Active Warps per SM [warp] | 48 | Block Limit Shared Mem [block] | 8 |
| Achieved Occupancy [%] | 68,50 | Block Limit Warps [block] | 6 |
| Achieved Active Warps Per SM [warp] | 32,88 | Block Limit SM [block] | 16 |

**Achieved Occupancy**
**Est. Local Speedup: 31.50%**

The difference between calculated theoretical (100.0%) and measured achieved occupancy (68.5%) can be the result of warp scheduling overheads or workload imbalances during the kernel execution. Load imbalances can occur between warps within a block as well as across blocks of the same kernel. See the CUDA Best Practices Guide for more details on optimizing occupancy.

Key Performance Indicators

## GPU and Memory Workload Distribution

Analysis of workload distribution in active cycles of SM, SMP, SMSP, L1 & L2 caches, and DRAM

| | | | |
|---|---|---|---|
| Average SM Active Cycles [cycle] | 17 028,40 | Average L1 Active Cycles [cycle] | 17 028,40 |
| Average L2 Active Cycles [cycle] | 14 547,25 | Average SMSP Active Cycles [cycle] | 16 160,08 |
| Average DRAM Active Cycles [cycle] | 50 160 | Total SM Elapsed Cycles [cycle] | 375 670 |
| Total L1 Elapsed Cycles [cycle] | 375 670 | Total L2 Elapsed Cycles [cycle] | 294 224 |
| Total SMSP Elapsed Cycles [cycle] | 1 502 680 | Total DRAM Elapsed Cycles [cycle] | 312 320 |

**SMSPs Workload Imbalance**
**Est. Speedup: 5.04%**

One or more SMSPs have a much higher number of active cycles than the average number of active cycles. Additionally, other SMSPs have a much lower number of active cycles than the average number of active cycles. Maximum instance value is 5.86% above the average, while the minimum instance value is 5.82% below the average.

Key Performance Indicators

To customize your report even further, you might want to learn about custom sections and writing your own rules.
You might also want to consider adding individual metrics.

You can ask Nsight Copilot questions about this report.